

Fig. 2: Actual-size PCB pattern of the front end for PC audio recording

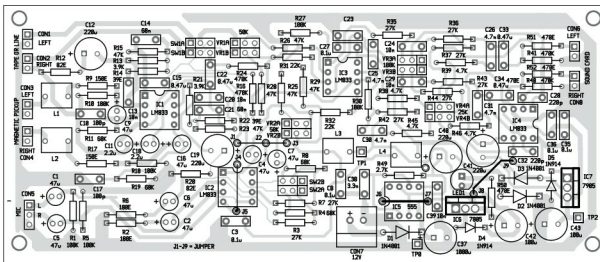


Fig. 3: Component layout of the PCB

Construction and testing

An actual-size, single-side PCB for the front end for PC audio recording is shown in Fig. 2 and its component layout in Fig. 3.

The unit is housed in a small sloping-front utility case made of plastic with most components mounted on a PC board measuring 157mm × 66mm approximately. The PCB may be mounted horizontally at bottom of the case. Mount pots, switches and LEDs on the front panel. Most connectors are single-hole RCA sockets, which need 6mm holes each. Mount these on the rear of the case. Connector CON7 for connecting 12V battery is also to be mounted on the rear of the case.

Since the case is predominantly

Test Points	
Test point	Details
TP1	+5.5V
TP2	-5.5V
(Voltage measured with respect to ground)	

plastic and provides very little shielding, a small shield plate cut from unetched PCB laminate is mounted under the main PCB, with its copper layer connected to signal earth.

With all passive parts fitted to the board, fit the three power supply ICs, inverting chopper IC5 and regulators IC6 and IC7. Then connect the 12V DC source. Check voltage +5.5V at the output pin of IC6 (TP1) and -5.5V at the output pin of IC7 (TP2) and mount four op-amp ICs (IC1-IC4)

in their respective bases.

Shielded audio cable should be used for wiring the microphone jack and magnetic pick-up input sockets. Also use this cable for wiring the tape/line input sockets, SW1, VR1 and VR2 to minimise susceptibility to noise and hum.

Solder a lead to the metal bodies of each of the pots on the front panel, with its other end

connected to signal earth via the earth lead from the LED. This should ensure good earthing of the control cases and front panel. Connect the left and right channels from CON6 and CON8 to the MIC input of the PC. Audacity software may be used for audio recording and for converting it to digital format.

EFY note. If you have an audio system such as a cassette tape recorder with Audio Out connector for headphones, you may skip this circuit. Just plug in a stereo jack into Audio Out and connect the other end of the audio cable to Mic In on your PC sound card. Launch Audacity software, play the cassette player and start audio recording by pressing Record. ●

Raj K. Gorkhali is a regular contributor to EFY